ENTRY 043   A Small Increment Is Too Noisy to Read — August 20, 2026 — Status: COMPLETE

Entry 043 — Checking Whether the GNN Has Plateaued, Before Grinding Toward 5,000

2,198 to 2,288 Circuits (+4%) Moved the Numbers the Wrong Way -- but That's Not the Same as a Real Plateau

1. Why This Check, Now

Growing the dataset toward 5,000 at full precision (Entry 042's methodology, no shot-count shortcuts) turned out to be much slower than expected -- roughly 10-20 circuits per batch round even after a free optimization (combining the 4 separate 4096-shot Aer calls into one 16,384-shot call, ~18% faster, statistically identical). Rather than keep grinding for the dozens of hours that would take, this entry asks the cheaper question first: is the GNN's improvement curve from Entries 041/042 still climbing, or did it already start to flatten with the 2,198 circuits already in hand.

2. The Result Is Genuinely Ambiguous

dataset size	GNN CV MAE	GNN CV R^2	floor-collapse MAE	floor-collapse n
1,376 (Entry 041)	2.64 pts	0.868	6.22 pts	70
2,198 (Entry 042)	2.04 pts	0.881	5.31 pts	112
2,288 (this entry)	2.17 pts	0.869	6.07 pts	117

The 1,376-to-2,198 jump (+60% more data) showed a clear, unambiguous improvement across every metric. The 2,198-to-2,288 jump (+4% more data) moved every metric in the WRONG direction -- but a 4% increment is far too small to distinguish a real plateau from ordinary training noise: each fold uses a different random 80/20 split and a freshly-initialized network, and with only ~23 floor-collapse cases per test fold, a handful of hard examples landing in one fold versus another can swing the reported MAE by more than this entire increment moved it.

Entry 043 — Conclusion

This entry does not answer whether the GNN has plateaued -- it answers that the question can't be answered from an increment this small, honestly reported rather than over-interpreted either direction. What it DOES confirm: the GNN remains dramatically ahead of v4.1 at every checkpoint (floor-collapse MAE 6.07 vs 14.67 points here, a 59% reduction, consistent with Entries 041 and 042), and R^2 has stayed in the same healthy 0.87-0.88 band across three retrainings. 2,288 circuits is a solid, validated working dataset -- not proven to be enough for a truly bigger GNN, but not shown to be insufficient either.

3. Next Step

Continue growing the dataset opportunistically in future sessions using the proven sequential, full-precision approach -- but without treating each small increment as a verdict on the model's ceiling. A real read on whether more of this kind of data still helps needs either a much bigger jump (the way 1,376 to 2,198 was) or averaging several training seeds per checkpoint to separate signal from noise, which Entry 043 did not have the compute budget to do this session.

Research Log — Index Addendum (Entry 043)

Entry	Title	Date	Status
Entry 043	Checking the GNN's Improvement Curve at 2,288 Circuits	Aug 20, 2026	Complete

Scripts: entry043_train_gnn.py. Data: quantumbridge_data/entry043_combined_dataset.json (2,288), quantumbridge_data/entry043_gnn_folds.json, quantumbridge_data/entry043_gnn_results.json, quantumbridge_data/entry043_growth_summary.json. Also includes a free, precision-preserving speedup to the Aer ground-truth pipeline in entry042_grow_parallel.py (one 16,384-shot call instead of four 4,096-shot calls).